

Topic 5: Alkenes

Related topics in Units 2, 4 and 5 will assume knowledge of this material.

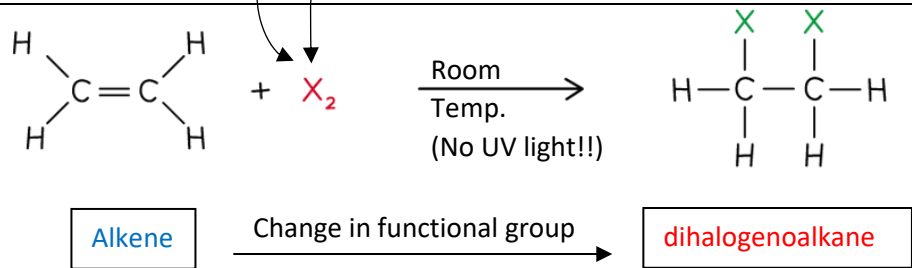
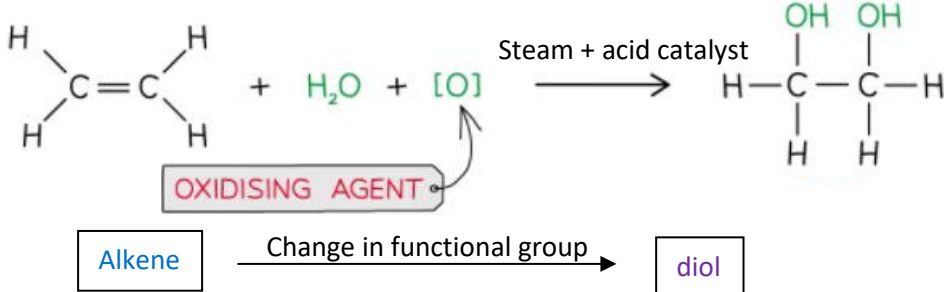
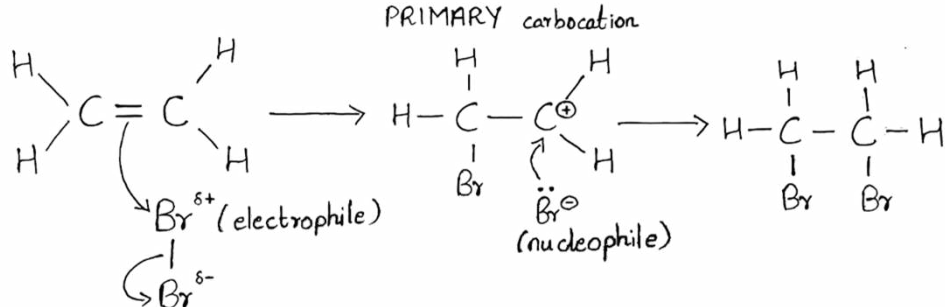
Students will be assessed on their ability to:

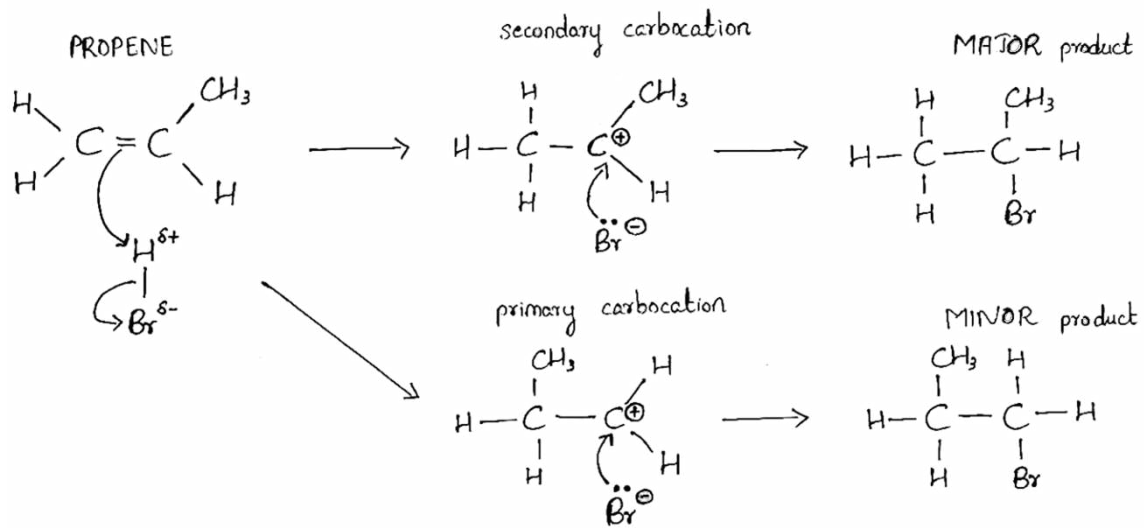
5.1	know the general formula of alkenes and understand that alkenes and cycloalkenes are hydrocarbons which are unsaturated (have a carbon-carbon double bond which consists of a σ bond and a π bond)
	All alkenes contain a C=C double bond so they are all unsaturated compounds General Formula: C_nH_{2n} The fact that they contain a C=C bond means they are more reactive than alkanes Cycloalkenes have the general formula $\text{C}_n\text{H}_{2n-2}$ When forming a covalent bond, the orbitals overlap in such a way to form two types of bonds: sigma and pi bonds π bonds (a.k.a C=C double bonds) are exposed and have high electron density hence they are more vulnerable to attack by electrophiles (electron-loving species) <u>Formation of a (σ) sigma bond</u> - Sigma bonds are formed from the end to end overlap of atomic orbitals (this is true for both s and p orbitals) - Rotation can occur around a sigma bond <u>Formation of a (π) pi bond</u> - <i>Pi bonds are formed from the sideways overlap of adjacent p orbitals</i> - A single pi bond has two-electron clouds, one above and one under the plane of the sigma bond (each electron cloud representing one bond containing two electrons) - This arrangement maximises the overlap of the p-orbitals
5.2	be able to explain geometric isomerism in terms of restricted rotation around a C=C double bond and the nature of the substituents on the carbon atoms
	In saturated compounds, the atoms/functional groups attached to the single, σ-bonded carbons are not fixed in their position due to the free rotation about the C-C σ-bond In unsaturated compounds, the groups attached to the C=C carbons can only be in one of two positions, and they remain fixed in that position

	For compound A and B to be geometric isomers, There must be two different atoms or group of atoms on each carbon of the double bond and there is restricted rotation about the double (C=C) bond This means each geometric isomer has different physical and chemical properties
5.3	understand the <i>E-Z</i> naming system for geometric isomers and why it is necessary to use this when the <i>cis-</i> and <i>trans-</i> naming system breaks down
E = Trans Z = Cis	<i>Cis-</i> and <i>trans-</i> naming system can only be used when we have only one functional group (or atom) on each side of the C=C double bond - If we have more than one, the <i>E-Z</i> naming system has to be used: - <i>Priority is decided by which atom has the higher atomic number</i> - If both atoms are above or below the double bond, then it is a Z-isomer - If one is above and one below, then it is an E-isomer
5.4	be able to describe the reactions of alkenes, limited to: - the addition of hydrogen, using a nickel catalyst, to form an alkane - the addition of steam, in the presence of an acid catalyst, to produce alcohols - the addition of hydrogen halides to produce mono-substituted halogenoalkanes - the addition of halogens to produce a di-substituted halogenoalkane - oxidation of the double bond by acidified potassium manganate(VII) to produce a diol
This is also a reduction reaction in terms of electron transfer Extensively used in manufacture of margarine	i An electrophilic addition reaction is the addition of an electrophile to a double bond (this includes all the reactions from i to iv) As stated above, the π bonds in alkenes are areas with high electron density and the fact that a π bond is weaker than a σ bond so less energy is needed to break π bond means it is more vulnerable (susceptible/ accessible) to electrophilic attack by electrophiles Alkene $\xrightarrow[\text{HEAT}]{\text{Pt/Ni CATALYST}}$ Alkane Change in functional group
	ii Alkene $\xrightarrow[(300^\circ\text{C} \ \& \ 60 \ \text{atm})]{\text{H}_3\text{PO}_4 \ \text{CATALYST, HEAT}}$ Alcohol Change in functional group
	iii Alkene $\xrightarrow[\text{TEMP.}]{\text{ROOM TEMP.}}$ Halogenoalkane Change in functional group

Dissolved in organic solvent

Heterolytic Fission in bond

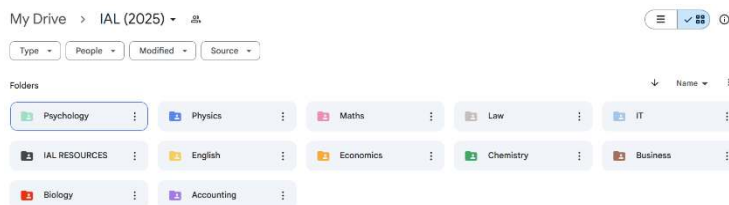
iv	 Alkene $\xrightarrow[\text{(No UV light!!)}]{\text{Room Temp.}}$ dihalogenoalkane Change in functional group
v	On top of electrophilic addition reaction, alkenes can also be oxidised by shaking it with cold dilute KMnO_4 (acidified potassium manganate(VII)) Pale purple solution turns colorless (a diol is produced) This can also be used as a test for alkenes just like bromine water  Alkene $\xrightarrow[\text{diol}]{\text{Change in functional group}}$
5.5	know the qualitative test for a $\text{C}=\text{C}$ double bond using bromine or bromine water
	<u>Bromine Water Test</u> Shake the unknown compound to be tested with bromine water in a test tube If it is unsaturated (has double bonds), addition reaction will occur and the solution will turn from orange to colorless (A qualitative test shows a yes or no, if its present or not, it doesn't how much it contains) It's better to know the colors of bromine in different states (orange in solution, red-brown liquid at room temperature and orange vapour)
5.6	be able to describe the mechanism (including diagrams), giving evidence where possible, of: - the electrophilic addition of bromine and hydrogen bromide to ethene - the electrophilic addition of hydrogen bromide to propene <i>Use of the curly arrow notation is expected – the curly arrows should start from either a bond or from a lone pair of electrons.</i> <i>Knowledge of the relative stability of primary, secondary and tertiary carbocation intermediates is expected.</i>
i	 PRIMARY carbocation - Although Br_2 in itself is non-polar, it can have an induced dipole when it is moved closer to a high electron density $\text{C}=\text{C}$ bond of the alkene

ii	 The diagram illustrates the electrophilic addition of HBr to propene. Propene reacts with HBr to form a secondary carbocation intermediate, which then reacts with Br⁻ to form the major product, 2-bromopropane. Alternatively, a primary carbocation intermediate is formed, which then reacts with Br⁻ to form the minor product, 1-bromopropane. - HBr is a polar molecule as Br is more electronegative than H, the H atom acts as an electrophile and is attracted to the electron-rich pi bond of propene and accepts a pair of electrons - The Br⁻ ion acts as a nucleophile - Secondary carbocations are more stable as there are more electron-releasing alkyl groups attached to the carbon with the positive charge - The major product is formed via the more stable carbocation intermediate - Order of stability for carbocations: Tertiary > Secondary > Primary
5.7	be able to describe the addition polymerisation of alkenes and draw the repeat unit given the monomer, and vice versa
	Addition polymerization is the reaction in which many monomers containing at least one C=C double bond form long chains of polymers as the only product
5.8	understand how chemists limit the problems caused by polymer disposal by: - i developing biodegradable polymers - ii removing toxic waste gases produced by the incineration of polymers
i	Biodegradable means it can be broken down by micro-organisms over time. The polyester and polyamide condensation polymers are considered to be biodegradable as they can be broken down using hydrolysis reactions, whereas poly(alkenes) cannot.
ii	The toxic waste gases produced by incineration, such as HCl, can be reacted with a base or carbonate. The base reacts with the acidic HCl, thereby neutralizing it.
	Further suggested practicals: - i investigating the difference in reactivity of alkanes and alkenes, including combustion, reaction with bromine water, reaction with acidified potassium manganate(VII) - ii preparation of cyclohexene from cyclohexanol - iii preparation of limonene from orange peel by steam distillation - iv preparation of Perspex® from methyl 2-methylpropenoate

Remarks

- Disclaimer: These Self-Study Booklet series are by no means intended to be a textbook replacement but instead are meant to be used alongside it
- This booklet is primarily exam-based and has been produced for last-minute revision in your exams by making the information in the syllabus into a simpler and more compact form factor
- **AND YOU ARE ADVISED TO USE THE LAB BOOK FOR CORE PRACTICALS AND FURTHER SUGGESTED PRACTICALS** which will not be covered in this booklet
- Thank you all for using this Self-Study Booklet and if you would like to find more resources like this one, do check out this google drive:
https://drive.google.com/drive/folders/1qpY-sGR9iinJbnquB53hx8pLk49pyxd1?usp=drive_link
- You will be redirected here:

(Note that there are only Self-Study Booklets/ CAQs for Physics and Chemistry)



- Do keep in mind that this is still a work-in-progress and you are welcome to add more resources to it- just drop a text to @aeth_en on discord!

Importance notice regarding Usage Rights

- There are certain images and illustrations used in this document from many different sources. The ownership of these images and/or illustrations belong to their rightful owners
- I do not own these images and/or illustrations
- Reprinting or Copying of this document and selling/ profiting from it is strictly prohibited
- This Self-Study Booklet is intended for students worldwide studying for the Edexcel International A-Levels and will remain free of charge

Help us improve

- This booklet is in its first edition so may contain mistakes here and there
- Please let us know of any room for improvement so that we can make it better!
- Contact: mth.2021education@gmail.com or @aeth_en on discord
- Best of luck in your A-Levels journey!!